Day 10 | Friday

November 13, 2026

10

Fraction operations by hand

9:00 AM–10:15 AM • Math • 75 min

Fractions in shop problems

10:30 AM–12:00 PM • Math • 90 min

Week 3 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Fraction operations by hand

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math • Math Week 3, Lesson 5

10:30 AM–12:00 PM • 90 min

Fractions in shop problems

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Math • Math Week 3, Lesson 6

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Fraction operations by hand

Today’s goal: Add, subtract, multiply and divide fractions on paper without a calculator and give every answer in lowest terms.

Name one use for today’s skill.

Textbook sections 4.6–4.7 pp.97–102

Exercise Book 4D Q1–10 (p.42)

Keep your notebook, textbook and exercise book ready. Exercise 4D is done by hand.

Textbook sections 4.6–4.7 pp.97–102

Recall and reconnect

Say the rule for each operation: a common denominator to add or subtract, improper fractions to multiply or divide.

Try first without displayed answers.

Resources: Exercise Book 4D Q1–10 (p.42)

Textbook sections 4.6–4.7 pp.97–102

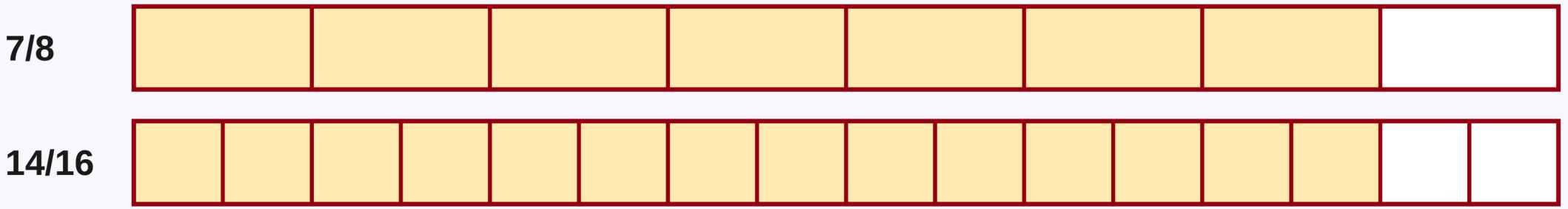
Add or subtract only when the parts match

1 Find a common denominator so both fractions count same-sized parts.

2 Rewrite each fraction, then add or subtract the numerators only.

3 Carry a full set into the wholes, or borrow one whole first.

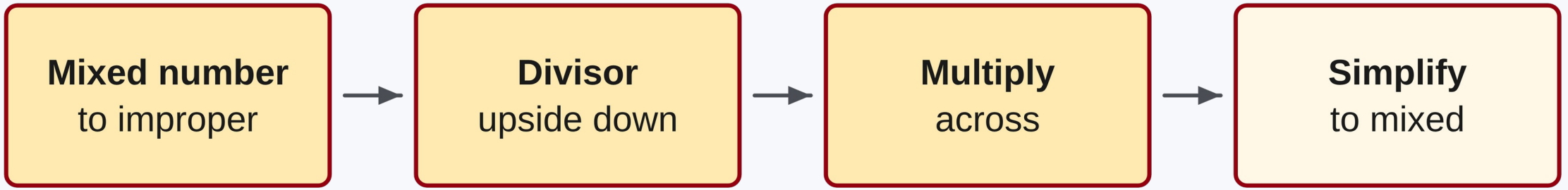
4 Write the answer in lowest terms.



Textbook sections 4.6–4.7 pp.97–102

Multiply and divide with improper fractions

1. Change each mixed number to an improper fraction first.
2. Multiply: numerator times numerator, denominator times denominator.
3. Divide: turn the divisor upside down, then multiply.
4. Simplify, then change back to a mixed number.



Textbook sections 4.6–4.7 pp.97–102

Estimate first, then trust the paper method

12 3/4 + 8 5/8 + 15 1/2 = 35 + 15/8 = 36 7/8 in

30 1/2 in ÷ 4 = 61/2 × 1/4 = 61/8 = 7 5/8 in

Estimate: about 13 + 9 + 16 = 38 in, close to 36 7/8 in.
Check: 4 × 7 5/8 in = 30 1/2 in.
Both are textbook examples, pages 98 and 101.



Together: 4D questions 2 and 9 by hand. One partner names the rule, the other writes the steps.

Textbook | page 98

Example 4.9 Total of a cut list

Find the total length of the three pieces: $12\frac{3}{4}$ in, $8\frac{5}{8}$ in, $15\frac{1}{2}$ in.

SOLUTION

1. Add the whole numbers: $12 + 8 + 15 = 35$.
2. Change the fractions to eighths and add them: $\frac{6}{8} + \frac{5}{8} + \frac{4}{8} = \frac{15}{8}$.
3. $\frac{15}{8} = 1\frac{7}{8}$. Carry the 1: $35 + 1 = 36$. The total is $36\frac{7}{8}$ in.

Textbook p.98 · Example 4.9: add mixed numbers, then carry the whole.

Textbook sections 4.6–4.7 pp.97–102

Textbook | page 101

For example, a $30\frac{1}{2}$ in bar cut into 4 equal pieces, ignoring the saw cuts, gives $30\frac{1}{2} \div 4 = \frac{61}{2} \times \frac{1}{4} = \frac{61}{8} = 7\frac{5}{8}$ in for each piece.

Textbook p.101 · Divide a mixed number: improper fraction, then the reciprocal.

Textbook sections 4.6–4.7 pp.97–102

Spot the mistake. Explain the correction.

A learner writes 2 1/2 × 1 1/2 = 2 1/4. What went wrong?

Discuss the method before checking the answer.

Wholes and fractions were multiplied separately. Use improper fractions: 5/2 × 3/2 = 15/4 = 3 3/4.

Textbook sections 4.6–4.7 pp.97–102

Work together

Solve 4D questions 2 and 9 by hand. One partner names the rule, the other writes the steps.

Work in pairs; alternate solver and checker.

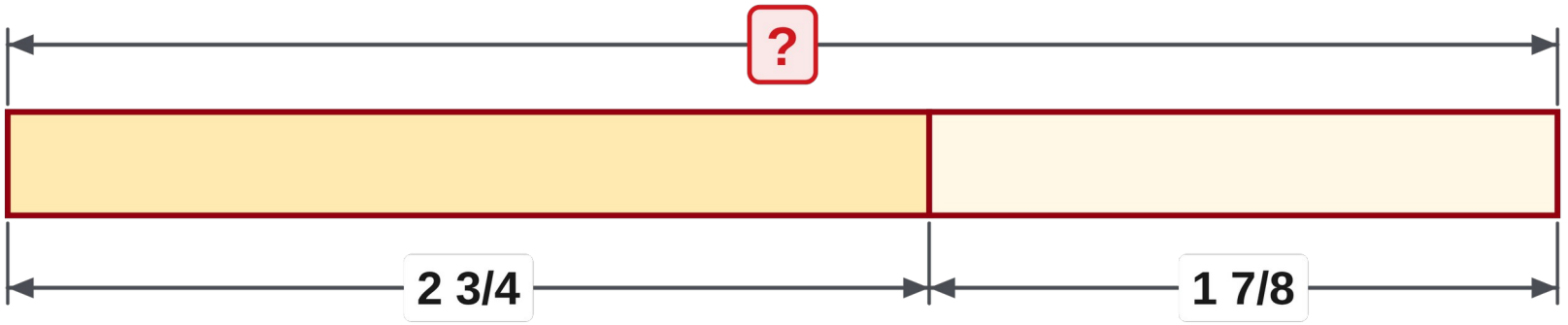
Resources: Exercise Book 4D Q1–10 (p.42)

Textbook sections 4.6–4.7 pp.97–102

Guided practice | 4D Q2, Q9

2 $2\frac{3}{4} + 1\frac{7}{8}$

9 $10\frac{1}{2} \div 1\frac{3}{4}$



Work with a partner. Take turns explaining the next step.

Textbook sections 4.6–4.7 pp.97–102

Compare your method | 4D Q2, Q9

2 $2\frac{3}{4} + 1\frac{7}{8}$

9 $10\frac{1}{2} \div 1\frac{3}{4}$

CHECK TOGETHER

Q2 4 5/8 Q9 6

Textbook sections 4.6–4.7 pp.97–102

Your scheduled practice

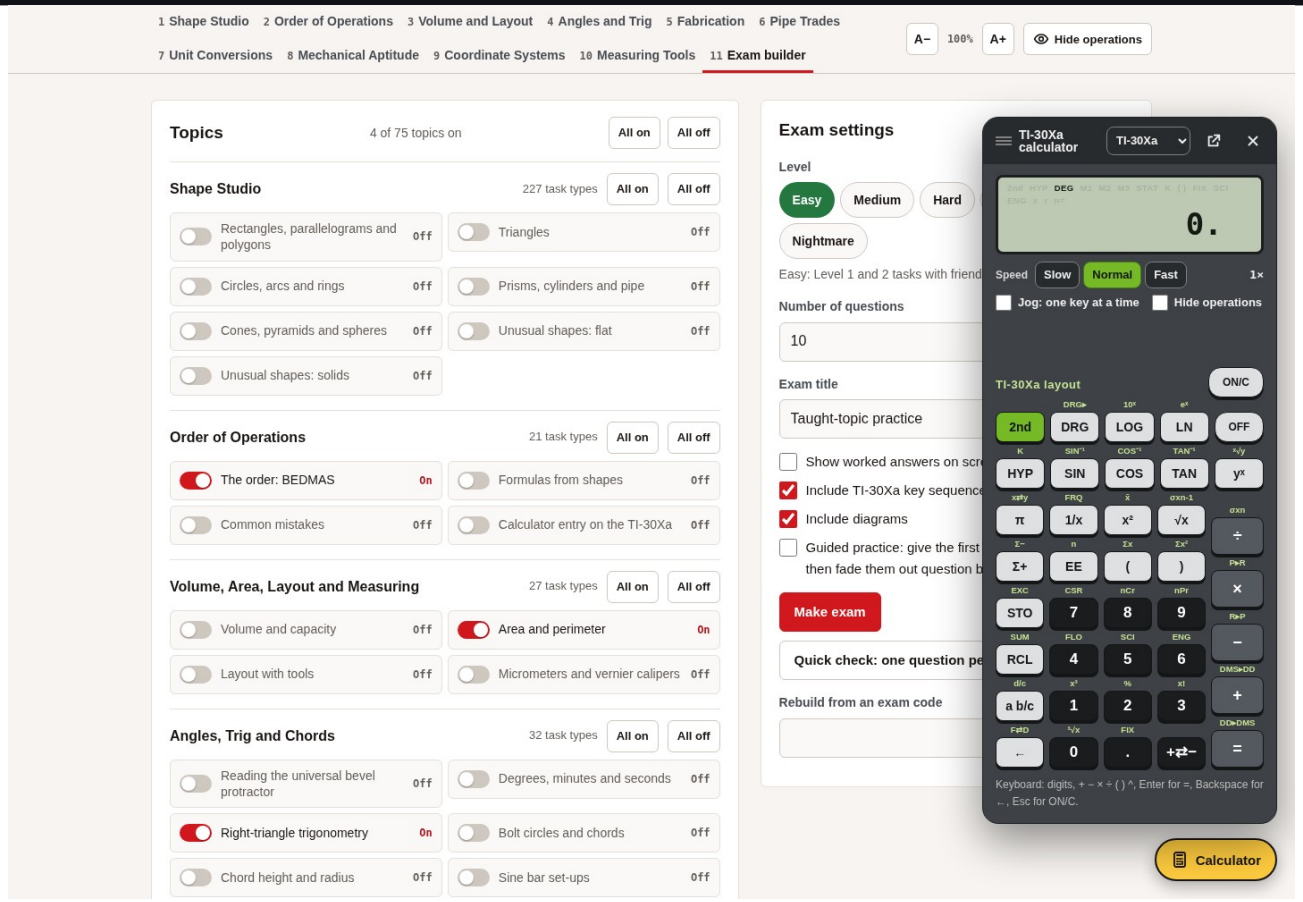
Complete 4D questions 1, 3, 4, 6, 7 and 8 on paper. Estimate first, give each answer in lowest terms, then check with the chosen calculator.

Show working and units; estimate, then check.

Resources: Exercise Book 4D Q1–10 (p.42)

Textbook sections 4.6–4.7 pp.97–102

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 4.6–4.7 pp.97–102

Practise by hand, then check in Numeracy Hub

1. Paper first: write every step of the assigned 4D questions, with no calculator.

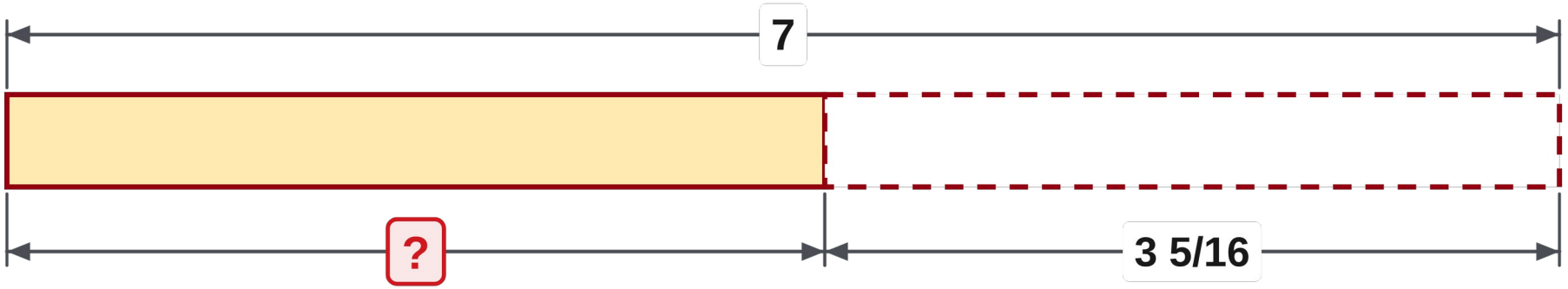
2. Name the rule for each question: a common denominator, or improper fractions.

3. Check afterwards: Numeracy Hub calculator, or The Basics fraction modules for a rule you missed.

Textbook sections 4.6–4.7 pp.97–102

Practise from the Exercise Book | 4D Q4

4 $7 - 3\frac{5}{16}$

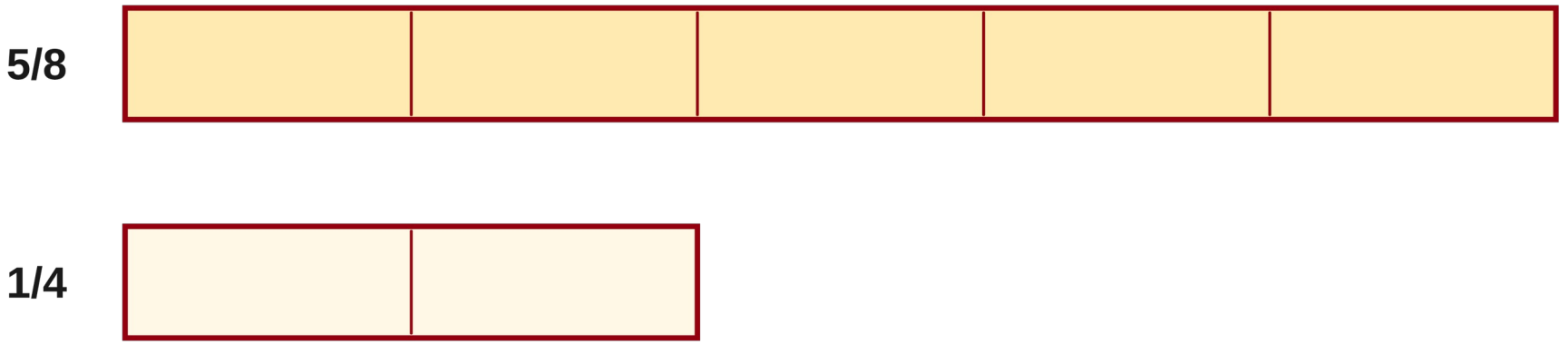


Write the operation, working and units before checking.

Textbook sections 4.6–4.7 pp.97–102

Practise from the Exercise Book | 4D Q8

8 $\frac{5}{8} \div \frac{1}{4}$



Write the operation, working and units before checking.

Textbook sections 4.6–4.7 pp.97–102

Check independently

Complete 4D questions 5 and 10 independently, without a calculator.

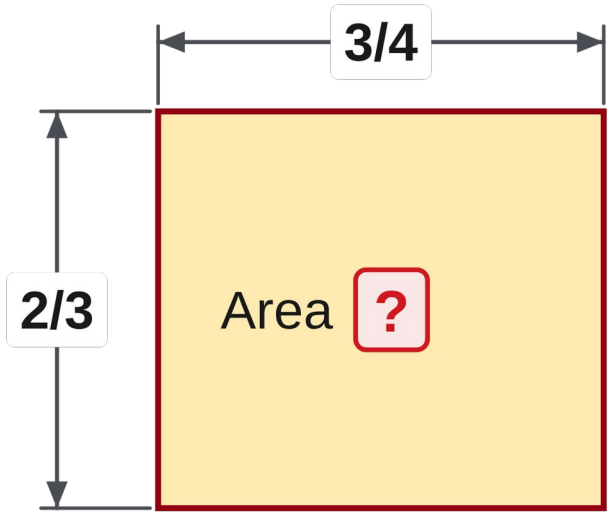
Answer independently and show the method.

Resources: Exercise Book 4D Q1–10 (p.42)

Textbook sections 4.6–4.7 pp.97–102

Independent check | 4D Q5

5 $\frac{3}{4} \times \frac{2}{3}$

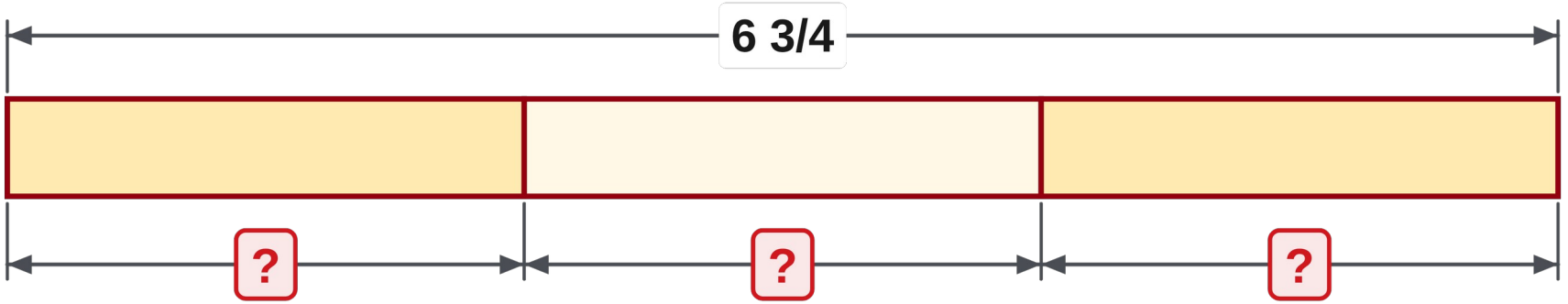


Answer independently. Keep worked solutions closed.

Textbook sections 4.6–4.7 pp.97–102

Independent check | 4D Q10

10 $6\frac{3}{4} \div 3$



Answer independently. Keep worked solutions closed.

Textbook sections 4.6–4.7 pp.97–102

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 4D Q1–10 (p.42)

Textbook sections 4.6–4.7 pp.97–102

Break

10:15 AM–10:30 AM

Return ready for: Fractions in shop problems

This break is excluded from teaching minutes.

Fractions in shop problems

Today’s goal: Choose the operation a shop problem needs, find a fraction of a length and round an equal part to the nearest 1/16 in.

Name one use for today’s skill.

Textbook sections 4.6–4.7 pp.97–102

Exercise Book 4B Q8–12 (p.38), 4C Q6–7 (p.41)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 4.6–4.7 pp.97–102

Recall and reconnect

Say what “of” means in “3/4 of a bar”, and how many sixteenths make one whole inch.

Try first without displayed answers.

Resources: Exercise Book 4B Q8–12 (p.38), 4C Q6–7 (p.41)

Textbook sections 4.6–4.7 pp.97–102

Read the job to choose the operation

1 Pieces joined or plates stacked: add.

2 Material removed, or a gap left: subtract.

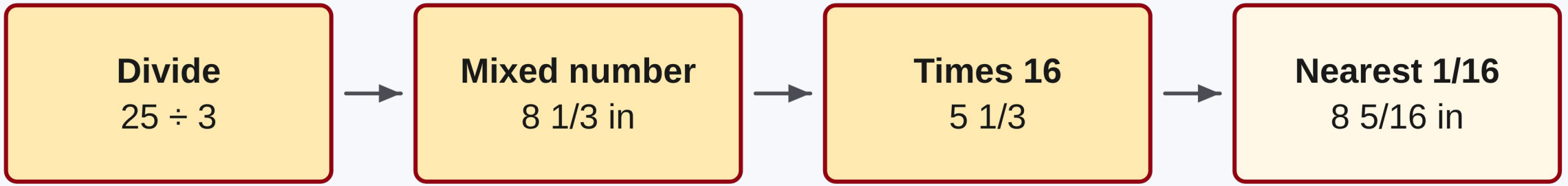
3 A fraction of a length, or repeated pieces: multiply.

4 A length shared into equal parts: divide.

Textbook sections 4.6–4.7 pp.97–102

Round lengths that are not on the tape

1. Divide: 25 ÷ 3 = 25/3 in.
2. Mixed number: 25/3 = 8 1/3 in.
3. Times 16: 1/3 × 16 = 5 1/3.
4. Round to whole sixteenths: mark 8 5/16 in.



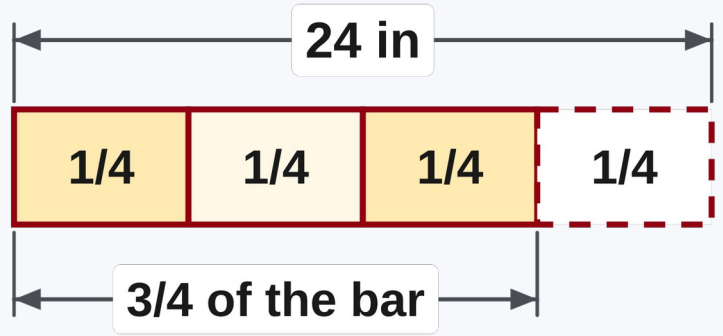
Textbook sections 4.6–4.7 pp.97–102

“Of” means multiply, not divide

Incorrect: 3/4 of a 24 in bar = 24 ÷ 3/4 = 32 in.
A part of the bar cannot be longer than the whole bar.

Correct: 3/4 × 24 = 72/4 = 18 in

Check with a sketch: three of four equal quarters.
Each quarter of 24 in is 6 in, and 3 × 6 = 18 in.



Together: 4B questions 8 and 10 with a labelled sketch. Name the operation before calculating.

Textbook | page 100

In word problems, the word "of" means multiply. $\frac{3}{4}$ of a 24 in bar is $\frac{3}{4} \times \frac{24}{1} = \frac{72}{4}$ = 18 in.

Textbook p.100 · In word problems, “of” means multiply.

Textbook sections 4.6–4.7 pp.97–102

Textbook | page 101

LENGTHS THAT ARE NOT ON THE TAPE

Some answers have denominators that are not on a tape, such as thirds. To mark them, multiply the fraction part by 16 and round to the nearest whole number of sixteenths. A 25 in bar cut into 3 equal pieces gives $\frac{25}{3} = 8\frac{1}{3}$ in. $\frac{1}{3} \times 16 = 5\frac{1}{3}$, which rounds to 5, so mark each piece at $8\frac{5}{16}$ in. Chapter 5 does the same job with decimal inches.

Textbook p.101 · Mark a length in thirds at the nearest sixteenth.

Textbook sections 4.6–4.7 pp.97–102

Spot the mistake. Explain the correction.

A 25 in bar is cut into 3 equal pieces. A learner marks each piece at 8 1/3 in. Can the tape show that mark?

Discuss the method before checking the answer.

No. Thirds are not on the tape. 1/3 × 16 = 5 1/3, so mark each piece at 8 5/16 in.

Textbook sections 4.6–4.7 pp.97–102

Work together

Solve 4B questions 8 and 10 with a labelled sketch. Name the operation before calculating.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 4B Q8–12 (p.38), 4C Q6–7 (p.41)

Textbook sections 4.6–4.7 pp.97–102

Guided practice | 4B Q8, Q10

8 A pipe has an outside diameter of $2\frac{3}{8}$ in and a wall $\frac{3}{16}$ in thick. What is the inside diameter?

10 A 2,400 mm length of channel is cut so one piece is $\frac{3}{4}$ of the length. How long is that piece?

Work with a partner. Take turns explaining the next step.

Textbook sections 4.6–4.7 pp.97–102

Compare your method | 4B Q8, Q10

8 A pipe has an outside diameter of $2\frac{3}{8}$ in and a wall $\frac{3}{16}$ in thick. What is the inside diameter?

10 A 2,400 mm length of channel is cut so one piece is $\frac{3}{4}$ of the length. How long is that piece?

CHECK TOGETHER

Q8 2 in Q10 1,800 mm

Textbook sections 4.6–4.7 pp.97–102

Your scheduled practice

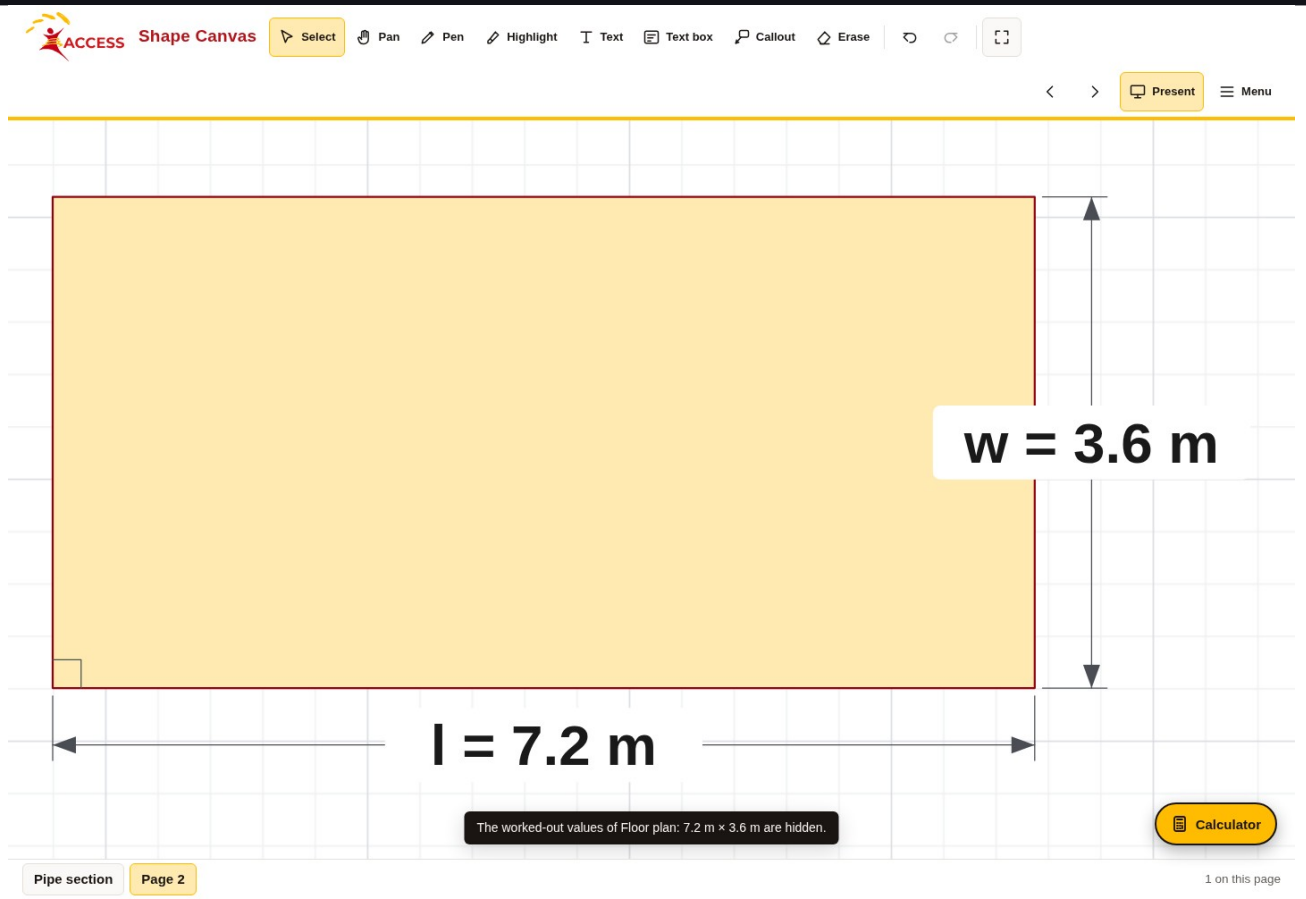
Complete 4B questions 9, 11 and 12, then 4C questions 6 and 7. Draw each part, name the operation and check one answer by working backwards.

Show working and units; estimate, then check.

Resources: Exercise Book 4B Q8–12 (p.38), 4C Q6–7 (p.41)

Textbook sections 4.6–4.7 pp.97–102

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 4.6–4.7 pp.97–102

Practise in Shape Canvas

1. Rebuild the 4C plate drawings in Shape Canvas with their labelled dimensions.

2. Hide the target value and work it on paper first.

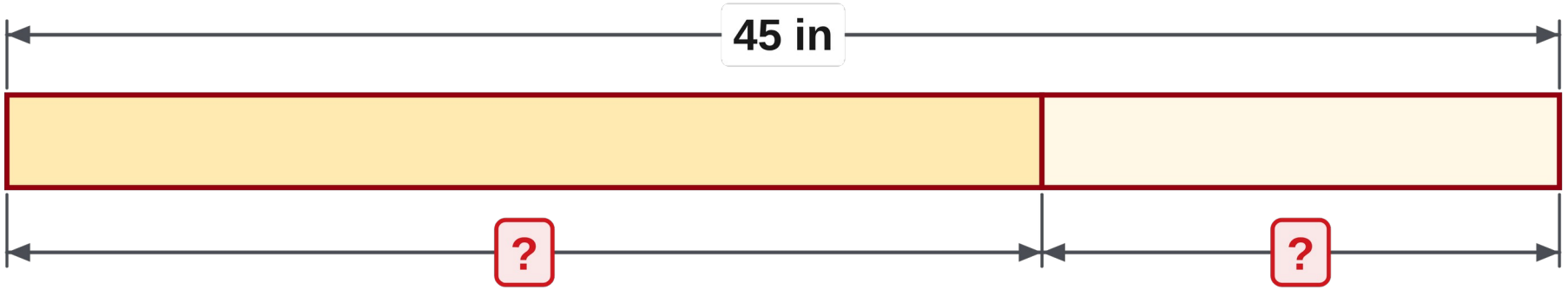
3. Check the exact fractions with the Numeracy Hub calculator after the paper attempt.

Textbook sections 4.6–4.7 pp.97–102

Practise from the Exercise Book | 4B Q11

A 45 in bar is cut so one piece is $\frac{2}{3}$ of the bar. How long are the two pieces?

Ignore the saw cut.

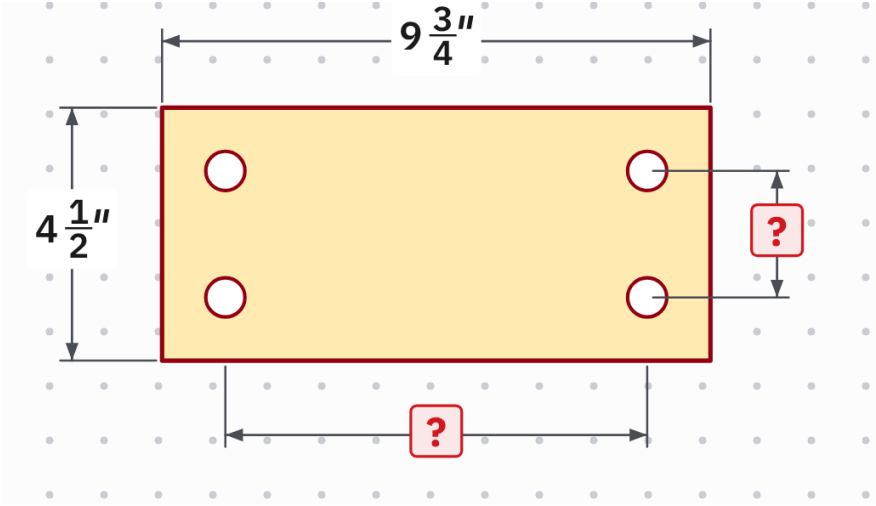


Write the operation, working and units before checking.

Textbook sections 4.6–4.7 pp.97–102

Practise from the Exercise Book | 4C Q7

Each hole centre is $1\frac{1}{8}$ in from two edges. Find the centre-to-centre distance along the plate and across it.



Write the operation, working and units before checking.

Textbook sections 4.6–4.7 pp.97–102

Check independently

Redo 4B question 12 and 4C question 6 independently. Give each length in lowest terms with its unit.

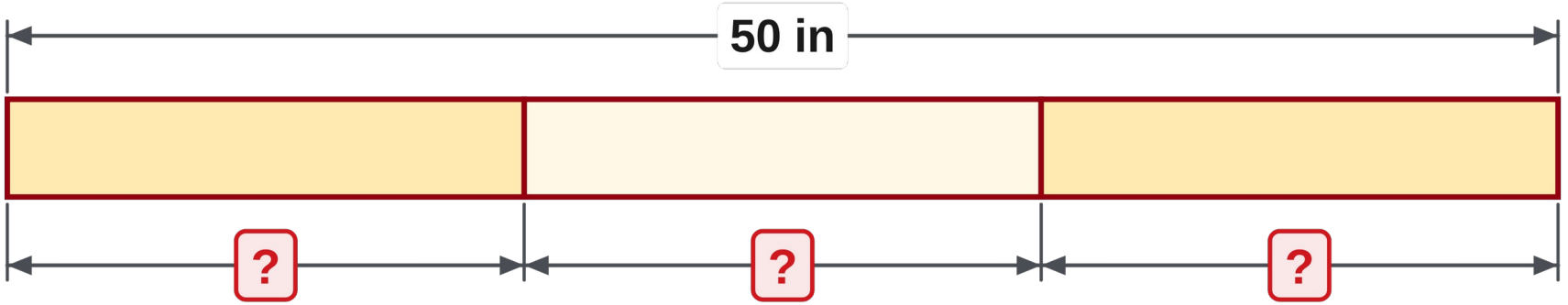
Answer independently and show the method.

Resources: Exercise Book 4B Q8–12 (p.38), 4C Q6–7 (p.41)

Textbook sections 4.6–4.7 pp.97–102

Independent check | 4B Q12

A 50 in bar is cut into 3 equal pieces. Ignore the saw cuts. How long is each piece, to the nearest $\frac{1}{16}$ in?

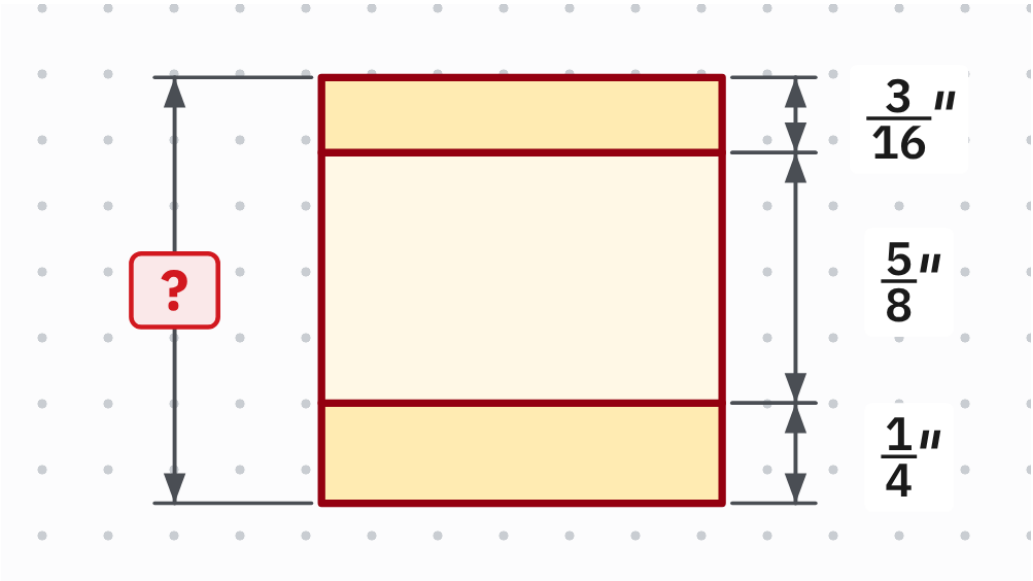


Answer independently. Keep worked solutions closed.

Textbook sections 4.6–4.7 pp.97–102

Independent check | 4C Q6

Three plates are stacked. How thick is the stack?



Answer independently. Keep worked solutions closed.

Textbook sections 4.6–4.7 pp.97–102

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 4B Q8–12 (p.38), 4C Q6–7 (p.41)

Textbook sections 4.6–4.7 pp.97–102